

INTUITION_OS_MASTER_SPEC.md

PART 2

PRODUCT MECHANICS

USER JOURNEYS · MENTOR SYSTEM · LEARNING ENGINE

Version 1.0

Core Product Loop

The entire product revolves around a single loop:

Problem → Attempt → Struggle → Mentor Interaction → Solution → Reflection → Evidence Extraction → Learning Memory Update → Better Future Mentorship

This loop is the heart of Intuition OS.

Every feature must strengthen this loop.

User Journey

Journey 1: Independent Solve

User opens LeetCode problem.

User solves without mentor assistance.

System tracks:

- Problem
- Duration
- Attempts
- Submission history

User reaches Accepted.

Reflection generated.

Evidence extracted.

Learning profile updated.

Why This Journey Matters

Many learning platforms only learn when users ask questions.

Intuition OS should learn from successful independent solves too.

Success without assistance is one of the strongest learning signals available.

Journey 2: User Gets Stuck

User opens problem.

User attempts solution.

User struggles.

Potential indicators:

- Long inactivity
- Multiple wrong submissions
- TLE
- Repeated failed approaches

System offers assistance.

Never forces assistance.

User chooses whether to engage.

Journey 3: Mentor Interaction

User requests help.

Mentor begins diagnostic phase.

Goal:

Understand user reasoning.

Not provide solution.

Mentor Philosophy

The mentor is not:

- ChatGPT
- Coding assistant
- Tutor

The mentor is:

A coach.

Coaches improve thinking.

They do not replace thinking.

Mentor Responsibilities

The mentor should:

- Diagnose
- Guide
- Encourage discovery
- Reinforce intuition

The mentor should not:

- Shortcut learning
 - Dump solutions
 - Reveal editorials
-

Mentor Escalation Model

Mentorship operates through increasing levels of assistance.

Level 1 — Reflection

Goal:

Understand reasoning.

Mentor asks:

- What have you considered?
- What seems difficult?
- What complexity does this create?

No hints.

No techniques.

No patterns.

Level 2 — Direction

Goal:

Provide orientation.

Examples:

- Consider whether ordering matters.
- Think about how information changes over time.
- What property remains consistent?

Still avoids revealing patterns.

Level 3 — Pattern Guidance

Goal:

Reveal conceptual direction.

Examples:

- This may benefit from maintaining a moving range.
- Think about techniques that reduce repeated work.

Pattern-level hints allowed.

Still avoids algorithms.

Level 4 — Strong Hint

Goal:

Unblock user.

Examples:

- Sliding Window may be useful.
- Consider maintaining frequencies.

Techniques allowed.

Detailed algorithms forbidden.

Level 5 — Solution Path

Only when:

- Explicitly requested
- User repeatedly fails
- User abandons coaching process

Provides:

- High-level solution path

Still avoids code unless explicitly requested.

Escalation Rules

Mentor should never skip levels.

User should gradually move upward.

This prevents accidental solution dumping.

Mentor Response Format

Every mentor response follows:

Observation Question Guidance

Example

Observation:

Current approach repeatedly compares elements.

Question:

What complexity does this create?

Guidance:

Can you think of a way to avoid repeated comparisons?

Why This Format Exists

Most AI assistants answer immediately.

The mentor should encourage reflection first.

Stuck Detection System

The system should never aggressively interrupt.

Users must retain control.

Stuck Signals

Possible indicators:

- 5-10 minutes inactivity
 - Multiple wrong answers
 - Repeated TLE
 - Repeated mentor requests
 - Editorial access
-

Mentor Prompt

Instead of:

"You seem stuck."

Use:

Need a nudge?

[Yes]

[Keep Working]

This preserves autonomy.

Learning Session

The most important domain object.

Everything derives from Learning Session.

Learning Session Definition

A Learning Session represents a complete attempt to solve a problem.

From opening the problem until completion or abandonment.

Learning Session Components

Problem Metadata

- Title
 - URL
 - Difficulty
 - Tags
-

Session Metadata

- Start time
 - End time
 - Duration
-

Submission History

- Submission count
 - Outcomes
 - Accepted status
-

Assistance History

- Mentor interactions
 - Hint levels
 - Editorial access
-

Reasoning History

- User-selected approaches
 - Mentor questions
 - User responses
-

Reflection

Generated summary.

Evidence

Extracted learning signals.

Why Learning Session Matters

It becomes the source of truth for:

- Reflections
 - Recommendations
 - Growth tracking
 - Personalization
-

Learning Memory System

Most AI tools remember conversations.

Intuition OS remembers learning.

Learning Memory Goal

Build longitudinal understanding.

Examples:

The mentor should know:

- What user struggles with
 - What user improves at
 - Which interventions work
-

Memory Hierarchy

Level 1

Current Session

Level 2

Recent Sessions

Last 10-20 problems.

Level 3

Topic History

Graphs

DP

Greedy

Arrays

Level 4

Behavior History

Persistence

Optimization

Pattern recognition

Level 5

Long-Term Learning Profile

Months of evidence.

Evidence Engine

The evidence engine converts events into learning signals.

This is the intellectual core of the product.

Inputs

Events:

- Problem opened
 - Hint requested
 - Editorial viewed
 - Submission made
 - Accepted
 - Reflection generated
-

Outputs

Evidence.

Examples:

- Solved Without Hint
 - Viewed Editorial
 - Recovered From Wrong Answer
 - Recovered From TLE
 - Needed Hint Level 2
 - Recognized Pattern
-

Critical Rule

Store evidence.

Do not store opinions.

Bad:

"User is intelligent."

Good:

"User solved 6 graph problems without hints."

Reflection Engine

The reflection engine transforms a session into learning insights.

Purpose

Not celebration.

Not gamification.

Learning reinforcement.

Reflection Structure

Session Summary

What happened?

Example:

You spent 35 minutes solving this problem.

Key Insight

What unlocked progress?

Example:

The breakthrough occurred when you recognized repeated comparisons were causing the bottleneck.

Behavioral Evidence

Observed behaviors.

Example:

- Solved independently
 - Recovered from TLE
 - Persisted through failure
-

Growth Impact

How profile changes.

Example:

This strengthens evidence for optimization reasoning.

Next Step

What should happen next?

Example:

Try another problem requiring brute-force optimization.

Reflection Quality Rules

Never invent learning.

Never exaggerate growth.

Every statement must trace back to evidence.

Confidence System

Every insight should have confidence.

High Confidence

Observed many times.

Medium Confidence

Observed multiple times.

Low Confidence

Observed once.

Mentor Memory Usage

The mentor should always use history.

Bad:

Use a hash map.

Good:

You solved four hash map problems this week without assistance.

Before using one here, what information do you think needs to be stored?

Core Product Moat

The moat is not:

- Better model
- Better UI
- Better prompts

The moat is:

Learning Memory

The accumulated history of how users think and improve.

Product Success Criteria

Users should eventually experience:

"I need fewer hints."

rather than:

"I received better hints."

That distinction defines success.

END OF PART 2